# Ravolexin Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-587/A

## Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions.

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. No changes to the approved analytical procedures are proposed in this submission. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control.

## Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Deviations observed during the campaign were investigated and closed with no impact to product quality. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. No changes to the approved analytical procedures are proposed in this submission. The control strategy links critical quality attributes to critical process parameters identified during development. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement.

## Facilities and Equipment

Manufacturing is performed at the approved commercial site using dedicated product-contact equipment trains. Equipment qualification and cleaning validation are maintained under the site validation master plan. No changes to the approved facility registration are proposed.

Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. No changes to the approved analytical procedures are proposed in this submission. Deviations observed during the campaign were investigated and closed with no impact to product quality. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection.

# Section 3.2.P.4 — Excipients

For the commercial formulation F-587/A, Kollidon 30 (Povidone, binder) is used. Plasdone K-29/32 was included in the excipient compatibility screen but was not carried forward into the commercial formulation. References to Kollidon 90F elsewhere in this dossier pertain to development history only.

## Basis for Selection

The selection of Kollidon 30 is justified by improved flowability during high-speed compression observed during development studies.

The excipient control strategy was developed in accordance with ICH Q6A, USP-NF and Ph. Eur. general monograph 2034.

## Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. No changes to the approved analytical procedures are proposed in this submission. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Cleaning verification swab results were below the health-based exposure limits derived for the product. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data.

## Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

## Facilities and Equipment

Manufacturing is performed at the approved commercial site using dedicated product-contact equipment trains. Equipment qualification and cleaning validation are maintained under the site validation master plan. No changes to the approved facility registration are proposed.

Cleaning verification swab results were below the health-based exposure limits derived for the product. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Statistical evaluation of batch release data demonstrates process capability well within specification limits. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement.

## Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. The control strategy links critical quality attributes to critical process parameters identified during development. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection.

## Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

Cleaning verification swab results were below the health-based exposure limits derived for the product. No changes to the approved analytical procedures are proposed in this submission. Deviations observed during the campaign were investigated and closed with no impact to product quality. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control.

The control strategy links critical quality attributes to critical process parameters identified during development. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions.

## Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Deviations observed during the campaign were investigated and closed with no impact to product quality. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data.

## Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. No changes to the approved analytical procedures are proposed in this submission. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation.

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation.

## Approved Drug Product Specification

The current approved commercial specification (SPEC-EX-1110 Rev 03) lists Povidone as Kollidon 30.

## Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Cleaning verification swab results were below the health-based exposure limits derived for the product. Training records for personnel involved in manufacturing and testing are maintained under the site quality system.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

## Container Closure System

The commercial packaging configuration consists of high-density polyethylene bottles with child-resistant polypropylene closures and induction seal liners. Container closure integrity was demonstrated by dye ingress testing. The packaging components comply with applicable food-contact regulations and compendial requirements for plastic packaging systems.

Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Deviations observed during the campaign were investigated and closed with no impact to product quality. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. No changes to the approved analytical procedures are proposed in this submission. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

## Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Cleaning verification swab results were below the health-based exposure limits derived for the product. Deviations observed during the campaign were investigated and closed with no impact to product quality. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results.

Statistical evaluation of batch release data demonstrates process capability well within specification limits. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement.

## Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Deviations observed during the campaign were investigated and closed with no impact to product quality. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles.

Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Statistical evaluation of batch release data demonstrates process capability well within specification limits. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. No changes to the approved analytical procedures are proposed in this submission.

## Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Cleaning verification swab results were below the health-based exposure limits derived for the product. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Deviations observed during the campaign were investigated and closed with no impact to product quality.

Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Deviations observed during the campaign were investigated and closed with no impact to product quality. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Cleaning verification swab results were below the health-based exposure limits derived for the product.

## Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

The control strategy links critical quality attributes to critical process parameters identified during development. Cleaning verification swab results were below the health-based exposure limits derived for the product. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Deviations observed during the campaign were investigated and closed with no impact to product quality. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions.

Cleaning verification swab results were below the health-based exposure limits derived for the product. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. No changes to the approved analytical procedures are proposed in this submission. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.